

Stage 3 — Head-Level Characterization

Proposal draft — for discussion

Phase A: Single-hop SIH Audit

September 2026

1 Stage 3 — Head-Level Characterization

1.1 Goal and scope

Stage 2 established that neither form of the Relation Index is a valid causal selection rule for single-hop in-context relations: the strongest causal heads have low or zero RI, the RI-selected heads are causally negligible, and the test-only name gap is *negatively* rank-correlated with causal importance, plausibly because the causal heads promote names by copying and the control penalizes copying. Stage 3 explains these findings at the resolution of the individual head. For every head in a fixed inventory it asks three questions:

1. **QK side:** what does the head attend to, and is the attention content-tracking or positional?
2. **OV side:** what does the head write — does its actual (contextual) output promote or suppress specific names?
3. **Taxonomy:** is the head an instance of a known type (induction, retrieval, function-vector, mover, negative mover), a definable new type, or unexplained?

Explicitly out of scope: inter-head edges, path patching, group ablations, minimal circuits, faithfulness, and validation families. Those are Stage 4. Two Stage-3 outputs feed Stage 4 directly: the copying verdict (experiment 1.3) and the per-position write profile (experiment 1.3).

1.2 Head inventory and groups

Group membership is fixed by the written rules below, computed from existing Stage-1/Stage-2 statistics, *before* any Stage-3 measurement. A head may satisfy several rules; it is assigned to every group it satisfies, and group overlaps are reported. All thresholds refer to already-saved quantities; no new selection is performed on Stage-3 data.

- **G1 (RI-high, causally weak).** Heads selected by either RI form but with small exact effect: the historical pair (L3H11, L9H22) and the test-only priority heads and name-gap top-10 members with $|I_h| < 0.1$ logits on the 178-pair extension (e.g. L11H4, L17H5).
- **G2 (causally strong, RI-low).** $|I_h| > 0.3$ logits (178-pair exact where available, otherwise 40-pair common subset) and outside every RI top-25: L17H1, L18H19, L27H6, L18H18, L21H18, L25H17, and the early writers L8H15, L14H23, L6H24 (include L9H16 here for its writer profile despite its name-gap origin).

- **G3 (concordant).** Heads with final-position QK-event agreement *and* $|I_h| > 0.3$: L16H21, L16H1, L22H5. This group is small; its near-emptiness is itself an audit result and must not be padded.
- **G4 (negative movers).** $I_h < -0.3$: L23H15, L19H16, L26H23.
- **G5 (random controls).** The 25 random extension heads of Stage 2, unchanged (seed 20260914). Every Stage-3 statistic is calibrated against this group; a fingerprint is reported as “present” for a head only when it exceeds the 95th percentile of G5.

1.3 Experiments

Experiments are ordered by cost: Block W uses weights only (no forward passes), Block D uses artifacts already saved by Stages 1–2, Block B requires new forward passes. Within a block the order is free.

Block W — weight-space analyses (no GPU)

W1. Name-copying and suppression score. For each head h , form the vocabulary-to-vocabulary map $C^h = W_E W_V^h W_O^h W_U$ restricted to the name-token rows used in our prompts (all first and last tokens of family names). Report (i) the *copying score*: the mean reciprocal rank of token t in row C_t^h over name tokens, and (ii) the *spectral sign*: the share of positive eigenvalues of C^h on the name subspace (Elhage et al. 2021 — positive eigenvalues indicate copying; strongly negative indicate suppression). *Evaluation*: this is the pre-registered test of the copy-penalizing hypothesis from Stage 2 §6.3. Prediction if the hypothesis is true: G2/G3 readers (layers 16–26) score high positive; G4 scores negative on the same subspace; G1 is indistinguishable from G5.

W2. Weight-based functionality scores (φ_R , saliency). Following Elhelo & Geva (2024): project each head as $M^h = \text{MLP}_0(\text{LN}(E)) W_{VO}^h W_U$ and compute the relation score φ_R = fraction of (child, mother) name pairs (s, t) with t in the top- k of row m_s^h , plus the suppression variant on $-m_s^h$ and per-token saliency $\sigma_t = \|e_t W_{VO}^h\| / \|e_t\|$. Because OLMo-2’s vocabulary ($\sim 100\text{k}$) differs from Llama-3.1, re-derive k and the classification threshold from their random-matrix baseline rather than reusing published values. *Evaluation*: a second, independent copying/relation classifier; disagreements between W1 and W2 are flagged, not averaged.

Block D — analyses of saved Stage-1/2 artifacts (no new forwards)

D1. Attention anatomy and the signed edge-following score. From the Stage-1 captured attention, classify each inventory head’s attention at (a) the final prompt position (the colon) and (b) test-fact positions, into: self, previous-token, fixed-offset, punctuation-anchored, or *content-tracking* (argmax follows the query-relevant fact’s tokens across the six variants/orders of a family). Then compute the methodology’s edge-following score in signed form: on corrupted twins, does the attention mass move to the *swapped* source, as content-tracking predicts, or stay at the same positions, as positional attention predicts? *Evaluation*: per head, a positional-vs-content label with a family-level consistency fraction; calibrate the content-tracking criterion on G5. Prediction: G1 labels positional (Stage 1 already showed this for L3H11, L9H22); the interesting unknown is G2’s readers.

D2. Promote/suppress index from stored logits. Stage 2’s Scope-F runs stored, per patched forward, the clean-answer logit, corrupted-answer logit and source-name logit. For each head define $P_h = \Delta\text{logit}(y_{\text{corr}})$ and $S_h = -\Delta\text{logit}(y_{\text{clean}})$ and place every head on the (P_h, S_h) plane, with the source-name change as a specificity axis. *Evaluation:* separates “promotes the retrieved name” (mover), “suppresses the alternative”, and “suppresses the retrieved name” (negative mover, IOI-style). Mostly a formalization of Stage 2 §7; the deliverable is one labeled scatter and per-head coordinates.

D3. Divergence-depth skeleton (methodology §6, item 6 — moved into Stage 3). Per layer ℓ , compute $d(\ell) = \|h_\ell^{\text{clean}} - h_\ell^{\text{corr}}\|$ at the answer position from cached runs. *Evaluation:* a layer-level consistency constraint for every head-level claim. Prediction: $d(\ell)$ jumps between layers ~ 11 and ~ 16 , matching the writer/reader split; a “critical” head above the completed divergence would be a contradiction to explain.

Block B — new measurements (GPU, single-head resolution)

B1. Contextual OV readout. For each inventory head, take its actual output at the colon on clean runs, $o_h = a_h W_O^h$ (its true contextual contribution to the residual stream), and project through the final LayerNorm and W_U (logit-lens caveat: report with and without final-LN folding). Report the rank and logit share of the gold answer, the distractor, and the source name. *Evaluation:* the contextual counterpart of the RI’s raw-embedding OV term. Prediction under the raw-embedding-miss mode (L27H6): contextual readout promotes the answer even where the raw-embedding score was zero.

B2. Known-type fingerprint battery. On synthetic sequences, independent of the kinship data: (i) *prefix-matching and copying scores* (Olsson et al. 2022) on random repeated token sequences; (ii) *retrieval score* (Wu et al. 2024): the fraction of copy events where the head’s top attention sits on the copied token, on needle-style prompts; (iii) *function-vector mediation* (Todd et al. 2024): add the head’s mean output over mother-of prompts to a zero-shot run and measure whether it induces the relation. *Evaluation:* all scores computed for all five groups; “has property” means above the 95th percentile of G5. Deliverable: the head $\times$ fingerprint matrix. This answers “new kind or re-labeled known kind?” per head.

B3. Per-position patch profile, Scope-F completion, and sufficiency. (i) For each inventory head, single-position noising patches across the test block: at which fact tokens does the effect enter? (ii) Complete the missing Scope-F measurements for L17H1, L27H6, L18H18. (iii) For the ten strongest heads, add the *denoising* direction (patch the clean activation into the corrupted run, all positions and final position): a sufficiency index alongside Stage 2’s necessity index. *Evaluation:* per head, a position profile (writer at fact tokens / reader at the colon / mixed) and a necessity–sufficiency pair (Heimersheim & Nanda 2024: the two directions answer different questions; carriers should show both, gates need not).

B4. Patchscopes readouts (discovery tool). For writer heads: patch the residual at their strongest write positions (from B3) into a token-identity few-shot prompt, sweeping source/target layers, to read what has been written there. For readers: a relation verbalization scope (“The mother of X is”) on the residual feeding the head. With fictitious names, any successful readout can only reflect in-context binding. *Evaluation:* free-text readouts are spot-check evidence only; control

every readout with the injection relevancy score $\Pr(\text{token} \mid \text{injection}) - \Pr(\text{token} \mid \text{no injection})$ (SelfIE), and never let a readout override a causal measurement.

Block N — novel techniques (ours)

N1. Contextual RI. Recompute the Relation Index on the saved Stage-1 events with one change: the OV term uses the head’s *actual residual-stream input* x_j^ℓ at its layer instead of the raw token embedding e_j . QK gate, controls, aggregation and thresholds stay exactly as published. *Evaluation:* report the rank correlation of contextual RI with exact importance next to the raw version’s (+0.135 historical, −0.25 test-only). This isolates how much of RI’s failure is attributable to the raw-embedding choice — a self-contained, publishable diagnosis of the definition, and the direct test of the “raw-embedding OV miss” mode (L27H6).

N2. Attention/value decoupling. For each strong head, two restricted patches at the colon: (i) transplant only the *attention pattern* from the corrupted twin while keeping clean values; (ii) transplant only the *values* while keeping the clean pattern. (Technique as in the IOI analysis; not yet applied here.) *Evaluation:* decomposes I_h into “attends to the right place” vs “writes the right thing” — at head resolution, without any circuit claim. Prediction: G3 heads (QK agrees, RI’s OV missed them) are pattern-dominated; pure copy-movers are value-dominated.

1.4 Evaluation framework and deliverables

1. **One matrix.** Every score (W1–N2) for every inventory head; thresholds calibrated on G5; overlaps between groups reported.
2. **Per-head profile pages** for G1–G4: attention anatomy, OV verdicts (raw, contextual, weight-space), fingerprint row, position profile, necessity/sufficiency, and a one-line classification: *known type / new type (named and defined) / unexplained*.
3. **Hypothesis verdicts.** (a) Copy-penalizing hypothesis: decided by W1 + B1 (+ B2 copying scores). (b) Raw-embedding-miss: decided by N1 + B1. (c) Writer/reader split: decided by B3 + D3.
4. **Completion criterion.** Stage 3 is complete when every G1–G4 head has a classification and hypotheses (a)–(c) each have a stated verdict with effect sizes. Heads classified “unexplained” are carried into Stage 4 as open items, not dropped.
5. **Discipline.** All analyses run on discovery families; the 87 validation families and the location world stay sealed. Bootstrap intervals are descriptive; no new significance claims are minted from Stage-3 exploration.